

DIGITAL LOGIC DESIGN

Number System Conversion

Practice Guide: Lectures 1 & 2

Name: _____ Date: _____ Score: _____ / 101

How to use this guide

1. Read the method box at the top of each part, then study the worked example (taken from the lectures).
2. Solve the problems by hand in the answer column, with no calculator. Use the margin for working.
3. Check your work against the answer key on page 10. If you get one wrong, redo it and find your mistake.
4. Do the parts in order. Each one builds on the one before it.

Roadmap

Part	Skill	Page	Questions
1	Any base to decimal (integers)	2	14
2	Decimal to binary, octal, hexadecimal (integers)	3	15
3	Decimal fractions to other bases	4	17
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Quick reference: 0 to 15 in every base

Decimal	Binary	Octal	Hex	Decimal	Binary	Octal	Hex
0	0000	0	0	8	1000	10	8
1	0001	1	1	9	1001	11	9
2	0010	2	2	10	1010	12	A
3	0011	3	3	11	1011	13	B
4	0100	4	4	12	1100	14	C
5	0101	5	5	13	1101	15	D
6	0110	6	6	14	1110	16	E
7	0111	7	7	15	1111	17	F

Part 1: Any Base → Decimal (Integers)

Method: the polynomial method

Value = Σ (digit $\times$ base^{position}). Number the positions from the right, starting at 0.

Step 1: write each digit with its position. Step 2: multiply the digit by base^{position}. Step 3: add the results.

Powers to know: 2: 1, 2, 4, 8, 16, 32, 64, 128, 256 | 8: 1, 8, 64, 512, 4096 | 16: 1, 16, 256, 4096. Hex letters: A=10, B=11, C=12, D=13, E=14, F=15.

Worked examples (from Lecture 1)

$$(11010)_2 = 1 \times 16 + 1 \times 8 + 0 \times 4 + 1 \times 2 + 0 \times 1 = (26)_{10}$$

$$(726)_8 = 7 \times 64 + 2 \times 8 + 6 \times 1 = 448 + 16 + 6 = (470)_{10}$$

$$(27FA)_{16} = 2 \times 4096 + 7 \times 256 + 15 \times 16 + 10 \times 1 = 8192 + 1792 + 240 + 10 = (10234)_{10}$$

Practice: convert each number to decimal

#	Convert	Your answer
1	$(1101)_2 \rightarrow$ base 10	
2	$(10110)_2 \rightarrow$ base 10	
3	$(111001)_2 \rightarrow$ base 10	
4	$(1010101)_2 \rightarrow$ base 10	
5	$(11111111)_2 \rightarrow$ base 10	
6	$(61)_8 \rightarrow$ base 10	
7	$(157)_8 \rightarrow$ base 10	
8	$(405)_8 \rightarrow$ base 10	
9	$(3042)_8 \rightarrow$ base 10	
10	$(2F)_{16} \rightarrow$ base 10	
11	$(B8)_{16} \rightarrow$ base 10	
12	$(3C5)_{16} \rightarrow$ base 10	
13	$(1A3F)_{16} \rightarrow$ base 10	
14	$(FF)_{16} \rightarrow$ base 10	

Part 2: Decimal → Binary, Octal, Hexadecimal

Method: repeated division

Divide the number by the new base. Write the quotient and the remainder. Keep dividing the quotient until it is 0. Read the remainders from bottom to top: the last remainder is the MSB and the first is the LSB.

For hexadecimal, remainders 10–15 are written as A–F.

Worked examples (from Lecture 2)

$(29)_{10}$: $29 \div 2 = 14 \text{ r } 1$ (LSB) · $14 \div 2 = 7 \text{ r } 0$ · $7 \div 2 = 3 \text{ r } 1$ · $3 \div 2 = 1 \text{ r } 1$ · $1 \div 2 = 0 \text{ r } 1$ (MSB) → **$(11101)_2$**

$(1341)_{10} = \mathbf{(10100111101)_2} = \mathbf{(2475)_8} = \mathbf{(53D)_{16}}$ (13 = D)

Practice

#	Convert	Your answer
1	$(13)_{10} \rightarrow \text{binary}$	
2	$(45)_{10} \rightarrow \text{binary}$	
3	$(92)_{10} \rightarrow \text{binary}$	
4	$(156)_{10} \rightarrow \text{binary}$	
5	$(255)_{10} \rightarrow \text{binary}$	
6	$(64)_{10} \rightarrow \text{octal}$	
7	$(100)_{10} \rightarrow \text{octal}$	
8	$(255)_{10} \rightarrow \text{octal}$	
9	$(511)_{10} \rightarrow \text{octal}$	
10	$(1000)_{10} \rightarrow \text{octal}$	
11	$(255)_{10} \rightarrow \text{hexadecimal}$	
12	$(500)_{10} \rightarrow \text{hexadecimal}$	
13	$(1000)_{10} \rightarrow \text{hexadecimal}$	
14	$(4095)_{10} \rightarrow \text{hexadecimal}$	
15	$(2026)_{10} \rightarrow \text{hexadecimal}$	

Part 3: Decimal Fractions → Binary, Octal, Hexadecimal

Method: repeated multiplication

Multiply only the fractional part by the new base. The whole-number part of the result is the next digit. Keep the new fractional part and repeat. Stop when it reaches 0 or when you have enough digits. Read the digits from top to bottom (first digit = MSB).

For a mixed number such as 374.37, convert the integer part by division and the fraction part by multiplication, then join them with the point.

Worked examples (from Lecture 2)

$(0.188)_{10} \times 2 \rightarrow 0.376 (0), 0.752 (0), 1.504 (1), 1.008 (1), 0.016 (0) \rightarrow (.00110)_2$ (5 digits)

$(0.23)_{10} \times 8 \rightarrow 1.84 (1), 6.72 (6), 5.76 (5) \rightarrow (0.165)_8$

$(374.37)_{10} \rightarrow \text{integer } 176, \text{ fraction } 0.5EB8 \rightarrow (176.5EB8)_{16}$

Practice

#	Convert	Your answer
1	$(0.625)_{10} \rightarrow \text{binary}$	
2	$(0.375)_{10} \rightarrow \text{binary}$	
3	$(0.8125)_{10} \rightarrow \text{binary}$	
4	$(0.1875)_{10} \rightarrow \text{binary}$	
5	$(0.3)_{10} \rightarrow \text{binary (5 digits)}$	
6	$(0.1)_{10} \rightarrow \text{binary (6 digits)}$	
7	$(0.5)_{10} \rightarrow \text{octal}$	
8	$(0.125)_{10} \rightarrow \text{octal}$	
9	$(0.4)_{10} \rightarrow \text{octal (3 digits)}$	
10	$(0.6)_{10} \rightarrow \text{octal (3 digits)}$	
11	$(0.75)_{10} \rightarrow \text{hexadecimal}$	
12	$(0.3125)_{10} \rightarrow \text{hexadecimal}$	
13	$(0.9)_{10} \rightarrow \text{hexadecimal (3 digits)}$	
14	$(0.2)_{10} \rightarrow \text{hexadecimal (4 digits)}$	
15	$(13.625)_{10} \rightarrow \text{binary}$	
16	$(25.75)_{10} \rightarrow \text{octal}$	
17	$(42.5)_{10} \rightarrow \text{hexadecimal}$	

Part 4: Fractions in Other Bases → Decimal

Method: negative powers after the point

Digits to the right of the point have negative positions: the first is base⁻¹, the next base⁻², and so on. Add the integer part and the fraction part.

Useful values: $2^{-1}=0.5$, $2^{-2}=0.25$, $2^{-3}=0.125$, $2^{-4}=0.0625$ | $8^{-1}=0.125$, $8^{-2}=0.015625$ | $16^{-1}=0.0625$, $16^{-2}=0.00390625$

Worked examples (from Lecture 2)

$(234.14)_8 = 2 \times 64 + 3 \times 8 + 4 \times 1 + 1 \times 0.125 + 4 \times 0.015625 = 156 + 0.125 + 0.0625 = (156.1875)_{10}$

$(4B3.3)_{16} = 4 \times 256 + 11 \times 16 + 3 \times 1 + 3 \times 0.0625 = 1024 + 176 + 3 + 0.1875 = (1203.1875)_{10}$

Practice: convert to decimal

#	Convert	Your answer
1	$(101.101)_2 \rightarrow \text{decimal}$	
2	$(11.011)_2 \rightarrow \text{decimal}$	
3	$(1101.1)_2 \rightarrow \text{decimal}$	
4	$(1010.0101)_2 \rightarrow \text{decimal}$	
5	$(0.111)_2 \rightarrow \text{decimal}$	
6	$(17.4)_8 \rightarrow \text{decimal}$	
7	$(62.2)_8 \rightarrow \text{decimal}$	
8	$(5.34)_8 \rightarrow \text{decimal}$	
9	$(100.01)_8 \rightarrow \text{decimal}$	
10	$(A.8)_{16} \rightarrow \text{decimal}$	
11	$(2F.C)_{16} \rightarrow \text{decimal}$	
12	$(1E.4)_{16} \rightarrow \text{decimal}$	
13	$(3.A8)_{16} \rightarrow \text{decimal}$	

Part 5: Binary ↔ Octal

Method: groups of 3 bits

Binary → Octal: split the bits into groups of 3. Start at the point and work outward: the integer part from the right, the fraction part from the left. Pad with zeros if a group is short. Convert each group (000–111 = 0–7).

Octal → Binary: replace each octal digit with its 3-bit group. Drop leading zeros at the end.

3-bit groups: 0=000 1=001 2=010 3=011 4=100 5=101 6=110 7=111

Worked examples (from Lecture 2)

$(10101)_2 \rightarrow 010 \mid 101 \rightarrow 2\ 5 = (25)_8$ $(25)_8 \rightarrow 010 \mid 101 = (10101)_2$

Practice

#	Convert	Your answer
1	$(101110)_2 \rightarrow \text{octal}$	
2	$(11010111)_2 \rightarrow \text{octal}$	
3	$(1111001)_2 \rightarrow \text{octal}$	
4	$(100101011)_2 \rightarrow \text{octal}$	
5	$(110101.101)_2 \rightarrow \text{octal}$	
6	$(47)_8 \rightarrow \text{binary}$	
7	$(725)_8 \rightarrow \text{binary}$	
8	$(1604)_8 \rightarrow \text{binary}$	
9	$(3.52)_8 \rightarrow \text{binary}$	
10	$(0.6)_8 \rightarrow \text{binary}$	

Part 6: Binary ↔ Hexadecimal

Method: groups of 4 bits

Binary → Hex: split the bits into groups of 4, working outward from the point (integer part from the right, fraction part from the left). Pad with zeros. Convert each group (0000–1111 = 0–F).

Hex → Binary: replace each hex digit with its 4-bit group.

Memory aid: 8-4-2-1. Example: $1011 = 8+0+2+1 = 11 = \text{B}$. The full 0–15 table is on page 1.

Worked examples (from Lecture 2)

$(10101)_2 \rightarrow 0001 \mid 0101 \rightarrow 1 \ 5 = (\mathbf{15})_{16}$ $(15)_{16} \rightarrow 0001 \mid 0101 = (\mathbf{10101})_2$

Practice

#	Convert	Your answer
1	$(11010110)_2 \rightarrow$ hexadecimal	
2	$(1011111001)_2 \rightarrow$ hexadecimal	
3	$(111100001111)_2 \rightarrow$ hexadecimal	
4	$(1101.1011)_2 \rightarrow$ hexadecimal	
5	$(101101.011)_2 \rightarrow$ hexadecimal	
6	$(3A)_{16} \rightarrow$ binary	
7	$(7F2)_{16} \rightarrow$ binary	
8	$(B09C)_{16} \rightarrow$ binary	
9	$(5.E8)_{16} \rightarrow$ binary	
10	$(2D.4)_{16} \rightarrow$ binary	

Part 7: Octal ↔ Hexadecimal and Fill-in Table

Method: use binary as the bridge

There is no direct shortcut between octal and hex. Convert the first number to binary (3 bits per octal digit or 4 bits per hex digit). Then regroup the same bits (4 bits for hex or 3 bits for octal) and convert.

Example: $(17)_8 \rightarrow 001\ 111 \rightarrow 0000\ 1111 \rightarrow (0F)_{16} = (F)_{16}$

A. Convert through binary

#	Convert	Your answer
1	$(275)_8 \rightarrow$ hexadecimal	
2	$(1234)_8 \rightarrow$ hexadecimal	
3	$(777)_8 \rightarrow$ hexadecimal	
4	$(3E)_{16} \rightarrow$ octal	
5	$(7C1)_{16} \rightarrow$ octal	
6	$(2F9)_{16} \rightarrow$ octal	

B. Complete the table (one value is given in each row)

Row	Decimal	Binary	Octal	Hexadecimal
1	217_{10}			
2		101101011_2		
3			345_8	
4				$9D_{16}$
5	1023_{10}			
6	50_{10}			

Part 8: Mixed Review, Error Spotting and Self-Check

A. Spot the mistake

#	Convert	Your answer
1	A student converts $(29)_{10}$ to binary and reads the remainders from top to bottom, getting $(10111)_2$. What went wrong? What is the correct answer?	Mistake / correct answer:
2	A student converts $(1101)_2$ to octal by grouping from the left: $11 \mid 01 \rightarrow (31)_8$. What went wrong? What is the correct answer?	Mistake / correct answer:
3	A student converts $(0.375)_{10}$ to binary by dividing by 2 and gets $(0.1875)_{10}$. Which operation should be used, and what is the correct binary result?	Mistake / correct answer:
4	A student says $(2F)_{16} = 2 \times 16 + 6 = 38$ because "F is 6". What is the correct value?	Mistake / correct answer:

B. Multi-base challenge

#	Convert	Your answer
5	$(1011011)_2$	Decimal: ____ Octal: ____ Hex: ____
6	$(3B)_{16}$	Decimal: ____ Binary: ____
7	$(400)_{10}$	Binary: ____ Octal: ____ Hex: ____
8	$(52.625)_{10}$	Binary: ____ Octal: ____ Hex: ____
9	$(111.011)_2$	Decimal: ____ Octal: ____ Hex: ____
10	$(7A.8)_{16}$	Decimal: ____ Binary: ____

C. Self-check: tick the skills you can do without notes

- Convert any base to decimal using positional powers (Part 1 and 4)
- Convert decimal integers to binary, octal and hex by repeated division (Part 2)
- Convert decimal fractions by repeated multiplication (Part 3)
- Convert between binary and octal or hex by grouping 3 or 4 bits (Parts 5 and 6)
- Convert octal $\leftrightarrow$ hex through binary (Part 7)

Answer Key

Check your answers once you have finished each part. Subscripts show the base. A fraction answer with a digit limit is rounded by cutting off, not by rounding.

Part 1 → decimal: 1) 13 2) 22 3) 57 4) 85 5) 255 6) 49 7) 111 8) 261 9) 1570 10) 47 11) 184 12) 965 13) 6719 14) 255

Part 2 → binary (1–5), octal (6–10), hex (11–15): 1) 1101_2 2) 101101_2 3) 1011100_2 4) 10011100_2 5) 11111111_2 6) 100_8 7) 144_8 8) 377_8 9) 777_8 10) 1750_8 11) FF_{16} 12) $1F4_{16}$ 13) $3E8_{16}$ 14) FFF_{16} 15) $7EA_{16}$

Part 3 → binary (1–6), octal (7–10), hex (11–14), mixed (15–17): 1) 0.101_2 2) 0.011_2 3) 0.1101_2 4) 0.0011_2 5) 0.01001_2 6) 0.000110_2 7) 0.4_8 8) 0.1_8 9) 0.314_8 10) 0.463_8 11) $0.C_{16}$ 12) 0.5_{16} 13) $0.E66_{16}$ 14) 0.3333_{16} 15) 1101.101_2 16) 31.6_8 17) $2A.8_{16}$

Part 4 → decimal: 1) 5.625 2) 3.375 3) 13.5 4) 10.3125 5) 0.875 6) 15.5 7) 50.25 8) 5.4375 9) 64.015625 10) 10.5 11) 47.75 12) 30.25 13) 3.65625

Part 5 → octal (1–5), binary (6–10): 1) 56_8 2) 327_8 3) 171_8 4) 453_8 5) 65.5_8 6) 100111_2 7) 111010101_2 8) 1110000100_2 9) 11.101010_2 10) 0.110_2

Part 6 → hex (1–5), binary (6–10): 1) $D6_{16}$ 2) $2F9_{16}$ 3) $F0F_{16}$ 4) $D.B_{16}$ 5) $2D.6_{16}$ 6) 111010_2 7) 11111110010_2 8) 1011000010011100_2 9) 101.11101000_2 10) 101101.0100_2

Part 7A → octal ↔ hex: 1) BD_{16} 2) $29C_{16}$ 3) $1FF_{16}$ 4) 76_8 5) 3701_8 6) 1371_8

Part 7B (table): 1) $217_{10} = 11011001_2 = 331_8 = D9_{16}$ 2) $363_{10} = 101101011_2 = 553_8 = 16B_{16}$ 3) $229_{10} = 11100101_2 = 345_8 = E5_{16}$ 4) $157_{10} = 10011101_2 = 235_8 = 9D_{16}$ 5) $1023_{10} = 111111111_2 = 1777_8 = 3FF_{16}$ 6) $50_{10} = 110010_2 = 62_8 = 32_{16}$

Part 8A (errors): 1) Read the remainders bottom to top: 11101_2 . 2) Group from the right: 15_8 . 3) Multiply by 2: 0.011_2 . 4) $F = 15$, so $2 \times 16 + 15 = 47$.

Part 8B (challenge): 5) Dec 91_{10} , Oct 133_8 , Hex $5B_{16}$ 6) Dec 59_{10} , Bin 111011_2 7) Bin 110010000_2 , Oct 620_8 , Hex 190_{16} 8) Bin 110100.101_2 , Oct 64.5_8 , Hex $34.A_{16}$ 9) Dec 7.375_{10} , Oct 7.3_8 , Hex 7.6_{16} 10) Dec 122.5_{10} , Bin 1111010.1000_2

Scoring guide

Count your correct answers across Parts 1–8. There are 101 questions in total (each table row and each challenge question counts as 1).

90% or more: excellent, you are ready for the exam. 70–89%: review the parts where you lost marks. Below 70%: re-read the method boxes and redo Parts 1–3 before moving on.